📘 Vim / Vi Command Notes

Vim (**Vi IMproved**) is a powerful text editor commonly used in Linux/Unix systems. It extends the original **vi** editor with more features. Vim is **mode-based**, meaning the same key can behave differently depending on the current mode.

This guide gives a **comprehensive summary of Vim commands**, with explanations and examples.

🕰️ Vim Modes

1. **Normal Mode (default)**
2. Used for navigation, editing, deleting, copying, and pasting.
3. Press `Esc` from any mode to return to Normal Mode.
4. **Insert Mode**
5. Used for typing text like a regular editor.
6. Entered from Normal Mode using: | Command | Action |
   |---------|--| | `i` | Insert before cursor | | `I` | Insert at beginning of line | | `a` | Append (insert after cursor) | | `A` | Append at end of line | | `o` | Open new line **below** and insert | | `O` | Open new line **above** and insert | | `cw` | Change word (delete + enter insert mode) |
7. Exit Insert Mode → Press `Esc`.
8. **Command-line Mode**
9. Used for saving, quitting, searching, and replacing.
10. Entered from Normal Mode with `:` (colon).
11. Examples: `:w`, `:q!`, `:wq`, `:%s/old/new/g`

🕰️ Basic Navigation (Normal Mode)

- `h` → move left
- `l` → move right
- `j` → move down
- `k` → move up

👉Think of them as arrow keys under your right hand.

Examples:

```
Hello World
```

- `h` moves cursor left - `l` moves cursor right - `j` moves cursor down - `k` moves cursor up

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🕰️Insert Mode Editing (special shortcuts)

While in Insert Mode: - `Backspace` → delete character left - `Ctrl-h` → same as Backspace - `Ctrl-w` → delete previous word - `Ctrl-u` → delete from cursor to beginning of line - `Enter` → new line below

👎 Movement keys like `h`, `j`, `k`, `l`, `w`, `b` **do not work** in Insert Mode. You must return to Normal Mode.

🕰️Save and Quit (Command-line Mode)

First press `Esc` to enter Normal Mode, then type: - `:w` → Save file - `:q` → Quit - `:wq` → Save and quit - `:q!` → Quit without saving - `:wq!` → Force save and quit

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🕰️Delete / Cut (Normal Mode)

- `x` → delete character under cursor
- `dw` → delete word after cursor
- `db` → delete word before cursor
- `dd` → delete current line
- `d$` or `D` → delete to end of line
- `d0` → delete to beginning of line
- `dG` → delete from cursor to end of file
- `dgg` → delete from cursor to beginning of file

Examples:

1. `x` (delete character)

```
Text: hello |world
Press: x
Result: hello orld
```

2. `dw` (delete word)

```
Text: hello |world again
Press: dw
Result: hello again
```

3. `dd` (delete line)

```
line one
|line two
line three

Press: dd
Result:
line one
line three
```

4. `d$` (delete to end of line)

```
Text: hello |world again
Press: d$
Result: hello
```

🕰️ Undo and Redo

- `u` → undo last change
- `Ctrl+r` → redo last undone change

🕰️ Copy (Yank) and Paste (Normal Mode)

- `yy` → copy current line
- `yw` → copy word from cursor
- `y$` → copy to end of line
- `yG` → copy to end of file
- `ygg` → copy to beginning of file
- `p` → paste after cursor (or below line if line was yanked)
- `P` → paste before cursor (or above line)

Cut is just Delete (`d`) + Paste: - `dd` → cut line - `dw` → cut word - `d$` → cut to end of line - Paste with `p` or `P`

🕰️ Search

- `/word` → search forward (down) for word
- `?word` → search backward (up) for word
- `n` → next occurrence
- `N` → previous occurrence

Example:

```
hello there
hello world
```

Type `/hello` + Enter → jumps to first "hello" Press `n` → jumps to next "hello"

🕰️Replace

- `rX` → replace character under cursor with `X`
- `:%s/old/new/g` → replace all `old` with `new`
- `:%s/old/new/gc` → replace all `old` with `new` (with confirmation)

🕰️Jump to Lines

- `G` (Shift + g) → last line of file
- `gg` → first line of file
- `:$` → last line (Command-line mode)
- `:1` → first line (Command-line mode)
- `:500` → jump to line 500
- `500G` → jump to line 500

🕰️Open Multiple Files

```
vim file1.txt file2.txt file3.txt
```

- `:next` → go to next file
- `:prev` → go to previous file
- `:args` → list files opened
- `:first` → go to first file
- `:last` → go to last file

👉If unsaved changes exist, Vim warns you. Use `:next!` or `:prev!` to force.

🕰️Split Windows

- `:split filename` → open file in horizontal split
- `:vsplit filename` → open file in vertical split
- `Ctrl+w w` → switch between splits

🕰️Quick Reference / Summary

- **Insert Mode** → `i`, `I`, `a`, `A`, `o`, `O`
- **Save/Quit** → `:w`, `:q`, `:wq`, `:q!`
- **Navigation** → `h`, `j`, `k`, `l`, `gg`, `G`
- **Delete** → `dd`, `dw`, `d$`, `d0`
- **Undo/Redo** → `u`, `Ctrl+r`
- **Search** → `/word`, `?word`, `n`, `N`
- **Replace** → `:%s/old/new/g`
- **Copy/Paste** → `yy`, `yw`, `y$`, `p`, `P`

👉**Golden Rule:** - `Esc` → Always return to **Normal Mode** - `:` → Enter **Command-line Mode** - `i`, `a`, `o` → Enter **Insert Mode**